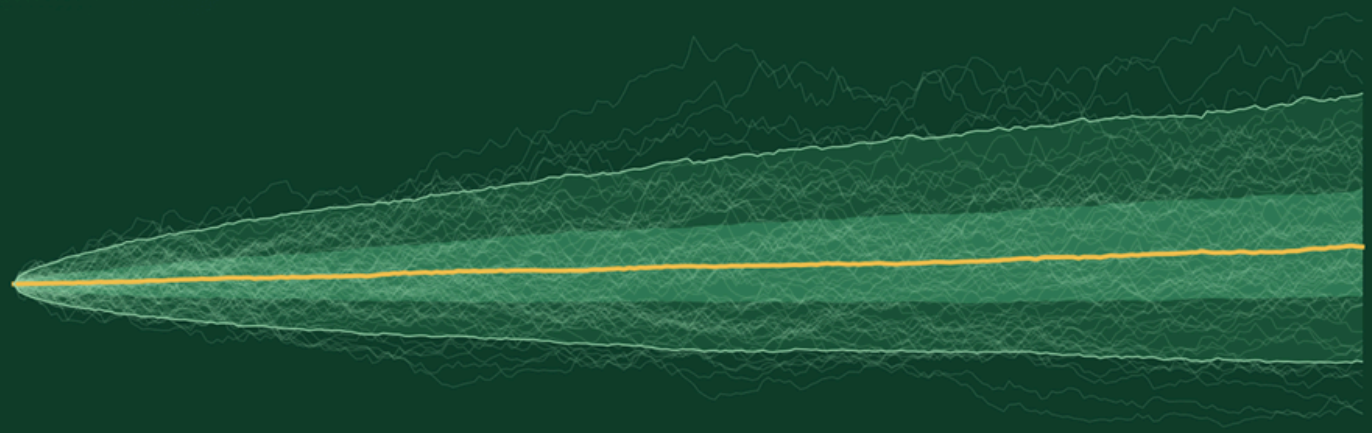


FOREIGN TRADE UNIVERSITY
FACULTY OF BANKING AND FINANCE

Applied Technology in Finance and Banking
PERSONAL PROJECT REPORT

Finfolio 2.0
Portfolio Attribution & Risk Analytics

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INDIVIDUAL ROLE	Product owner/ Financial logic lead
GROUP	G09 - K63
COURSE	Information Technology Applications in Finance and Banking (NHA408E)
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Table of Contents

Introduction and Individual Footprint 2

Distinctive Contributions 3

Development Process and Collaboration 5

Evidence and Verification 6

Limitations and Lessons Learned 7

Finfolio 2.0

Introduction and individual footprint



FinFolio 1.0 already allowed users to calculate portfolio returns, inspect risk indicators, compare results with a benchmark and explore alternative weights. I therefore began by examining what the existing product could not explain clearly. Its investment coverage was relatively narrow, the dashboard contained many numbers in one place, and it did not fully show how individual holdings contributed to portfolio return or risk.

I helped frame FinFolio 2.0 around a practical question: after an investor enters a portfolio, can the application explain what happened, which holdings or groups produced the result, and where the main risks lie? We chose self-directed retail investors and finance students with basic financial knowledge as the main users. They need a way to review a small multi-asset portfolio, not a trading platform or a source of automatic buy-and-sell advice.

→ As Product Owner, I worked on the target user, core workflow and boundaries of the product. As Financial Logic Lead, I defined the meaning of the portfolio measures, developed the attribution methods, and set out how to check that the numbers agreed with one another. The expected sequence was to enter and validate holdings, calculate portfolio value, explain return and risk, compare with a benchmark, and only then assess a possible change in allocation.

MY ACCOUNTABILITY	WHAT I DELIVERED
Product direction	Problem selection, target user, product boundaries and a clear analysis-to-decision journey.
Financial methodology	Return contribution; MCR/Euler; FX return and risk decomposition; optimization objectives and constraints, benchmark comparability,
Attribution architecture	Six grouping lenses, category classification principles, aggregation rules and reconciliation identities.
Cross-team communication	Financial specifications, acceptance rules and explanations for QA, UI/UX and final presentation.

CEO Company

Distinctive contributions

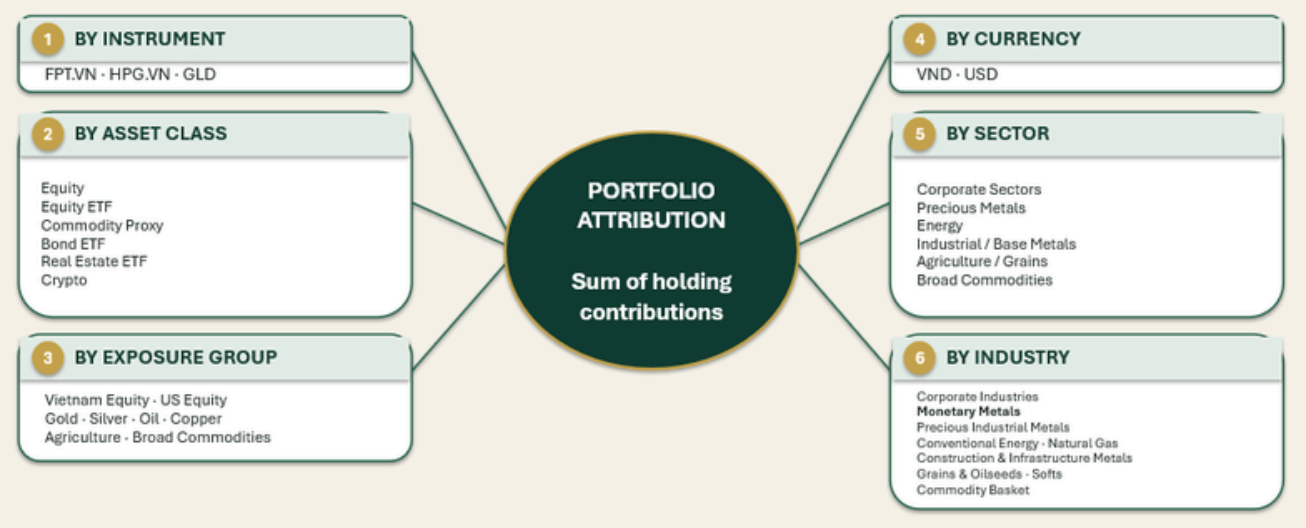
1 Portfolio value and return contribution

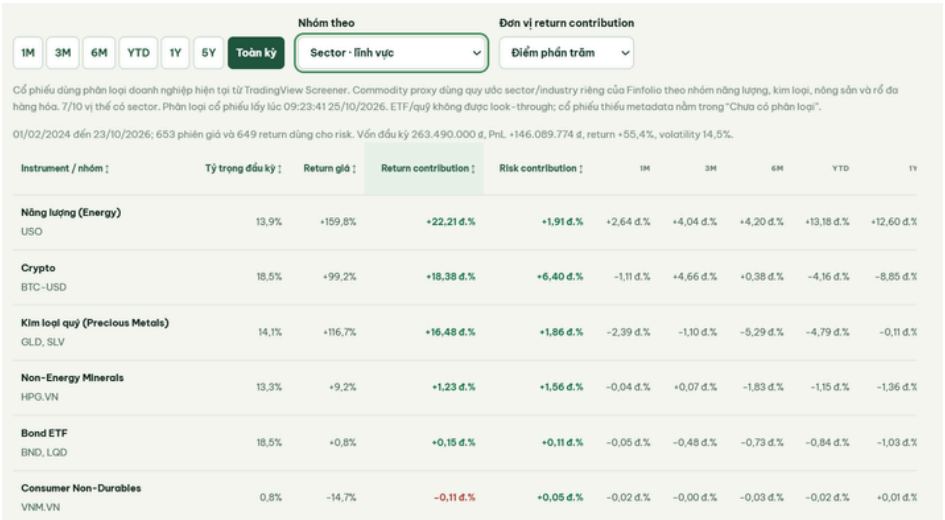
The user supplies the initial amount invested in each holding, in VND or USD. After any required currency conversion, the application calculates the quantity held at the first valid observation and keeps that quantity fixed. This is a buy-and-hold calculation, so weights are allowed to move as prices change. Moreover, I also search for ratios of return to analyze and prpesent for user. I decide what ratios are on the main dashboard, while more complicated ratios will be on an add on sections that user might choose to read more (the comprehensive ratio tables arre presented in group sldie deck)

$$Portfolio\ value\ at\ t\ =\ \Sigma\ quantity_i \times price_i,t \times FX_i,t$$

$$Return\ contribution_i\ =\ starting\ weight_i \times holding\ return_i$$

2 Multi-dimensional attribution





Designing the grouping system was a substantial part of my individual work. A table of contributions by ticker can identify the strongest and weakest positions, but it does not show whether the result is concentrated in a wider exposure. I proposed six ways to view the same calculation, moving between individual holdings and broader categories without changing the portfolio total

Distinctive contributions

3 Foreign-exchange attribution

When an investor holds a USD-denominated instrument but measures the portfolio in VND, the final return has two main sources: the instrument's local-currency price and the USD/VND exchange rate. I designed a decomposition that makes these sources visible, rather than showing only the combined VND return.

$$VND\ return = local\ price\ return + FX\ return + interaction$$

$$Interaction = local\ price\ return \times FX\ return$$

Giá tài sản và tỷ giá đóng góp thế nào?

Return: 01/02/2024 đến 23/10/2026 · cùng kỳ chọn ở bảng đóng góp. Risk: toàn kỳ 01/02/2024 đến 23/10/2026.

MÃ / ĐỒNG NIÊM YẾT	GIÁ TÀI SẢN	TỶ GIÁ USD/VND	TƯƠNG TÁC GIÁ × TỶ GIÁ	TỔNG ĐÓNG GÓP
FPT.VN · VND	−2,58 đ.%	+0,00 đ.%	+−0,00 đ.%	−2,58 đ.%
HPG.VN · VND	+1,23 đ.%	+0,00 đ.%	+0,00 đ.%	+1,23 đ.%
GLD · USD	+8,78 đ.%	+0,55 đ.%	+0,51 đ.%	+9,85 đ.%
BND · USD	−0,31 đ.%	+0,54 đ.%	−0,02 đ.%	+0,21 đ.%

4 Portfolio risk measures and risk attribution

I work on to choose suitable risk metrics for the website, which I fully present in the slide deck. Moreover, I have to think about how to measure risk when attribute assets to different groups. I selected the Euler method to divide estimated portfolio volatility into component contributions. Marginal Contribution to Risk (MCR) measures how estimated volatility responds to a small change in one holding's weight. Euler Risk Contribution multiplies that sensitivity by the holding's current weight. These two measures are connected parts of one calculation, rather than separate risk models

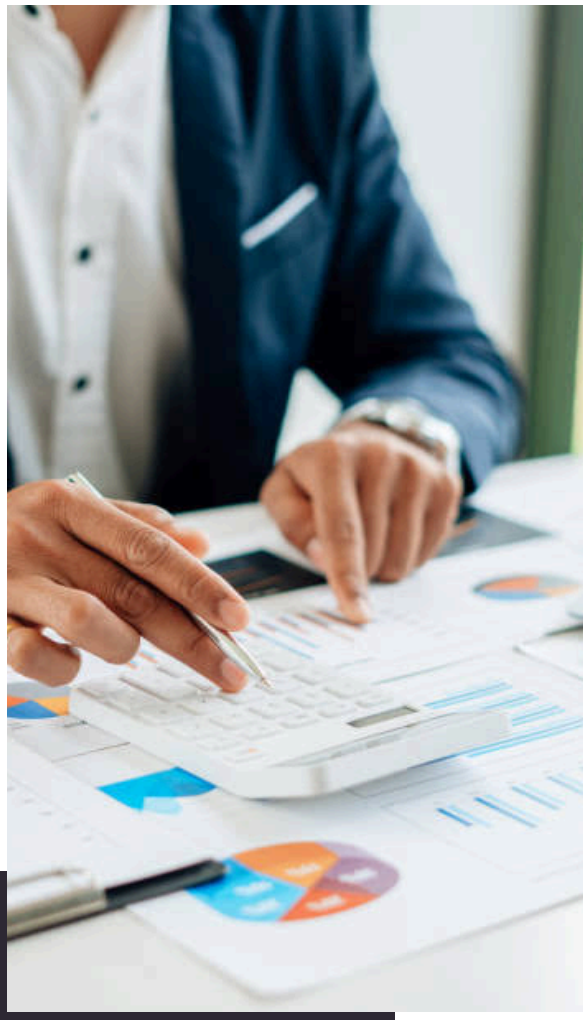
$$Volatility_p = \sqrt{(252 \times w^T \Sigma w)}$$

$$MCR_i = \sqrt{252} \times (\Sigma w)_i / \sqrt{(w^T \Sigma w)} \qquad RC_i = w_i \times MCR_i$$

5 Optimization assumptions

FinFolio 1.0 already used Markowitz-style allocation objectives. My role in 2.0 was not to invent a new optimization model or program the numerical solver. I set out how the objectives should be defined and presented, which assumptions should be visible, and how to compare the current portfolio with the suggested reference weights fairly.

Development process and collaboration



Week 1

we considered several ways to extend FinFolio 1.0. I argued that attribution should be central because it answered a question that the existing metrics left unresolved: which holdings were responsible for the portfolio result? I helped define the target audience, choose a practical scope, and organise the flow around explaining an existing portfolio before proposing changes to it.

Week 2-4

I had translated this direction into a set of calculation rules and review conditions. For return attribution, the parts had to equal the whole. For Euler risk, the holding contributions had to sum to estimated volatility. The FX components had to reproduce the VND return, while optimized allocations had to respect the stated caps. These checks gave the other workstreams clear targets beyond simply displaying a plausible number

Team member / area	What they owned	How my work connected
Lê Bảo An / Development	Financial calculation modules, optimization, simulation and deployment	I provided the intended meaning, objectives, limitations and numerical checks.
Trần Minh Ngọc / Data	Sources, currency metadata, benchmark selection and common-date alignment	I specified what the return and risk calculations required from the data.
Nguyễn Quỳnh Anh / QA	Acceptance cases, regression tests and workflow checks	We turned financial equalities and invalid-input conditions into checks.
Nguyễn Ngọc Anh / UI/UX	Navigation, information hierarchy and visual presentation	I reviewed whether labels and explanations matched the finance method.

As the product moved towards the final demonstration, I reviewed the explanations alongside the calculations, and give feedbacks to coder about any mistakes or omitted assumptions that I found

Evidence and verification

My contribution to FinFolio 2.0 can be verified through the project's methodology documents, the final website, and the financial calculations I can explain and reproduce independently.

As Product Owner and Financial Logic Lead, my main responsibility was to establish how the application should evaluate a portfolio and communicate its results. This involved defining the financial methods, developing the attribution framework, setting calculation requirements, and reviewing whether the outputs were consistent with their intended financial meaning.

The methodology documents record these decisions, while the final product demonstrates how they were implemented by the development team. The following table summarises the main evidence of my work.

Area of Contribution	Supporting Evidence	What Can Be Verified
Product Direction & Scope	docs/WEEK4.md; website workflow	Definition of the target users, core financial questions, product scope, and distinction between portfolio analysis and investment advice
Portfolio Value & Return Logic	docs/WEEK4.md; demo/finance.mjs	Buy-and-hold valuation, VND-based portfolio returns, and consistency between portfolio value and total return
Return Attribution Framework	demo/attribution.mjs; Analysis page	Asset-level return contribution, calculation over different periods, and reconciliation with total portfolio return
Multi-dimensional Attribution	demo/attribution.mjs	Classification into six dimensions and aggregation of contributions.
Marginal & Euler Risk Contribution	docs/WEEK4.md; demo/attribution.mjs	Use of covariance and portfolio weights to calculate risk contributions and reconcile them with total estimated volatility
FX Attribution	docs/FX_ATTRIBUTION.md; Analysis page	Decomposition of foreign-asset returns into local-price, exchange-rate, and interaction effects
Optimization Framework	docs/OPTIMIZATION.md; Optimization page	Definition of optimization objectives, allocation constraints, historical assumptions, and interpretation of reference allocations
Financial Metrics & Interpretation	docs/FINAL_METHODS.md; Advanced Metrics	Metric definitions, sample requirements, calculation conventions, and explanations of results and limitations

Limitation and lesson learned

Limitation

First, the analysis does not fully capture real investment performance.

The current portfolio valuation framework mainly relies on historical price returns. It does not fully account for dividend reinvestment, transaction fees, taxes, additional deposits or withdrawals, or the timing of actual trades. As a result, the performance displayed by FinFolio should be understood as the historical performance of a hypothetical portfolio under the model's assumptions, rather than the exact realised profit or loss of an investor.

Second, differences in trading calendars create additional challenges for multi-asset portfolios.

One of the complexities of expanding FinFolio beyond Vietnamese equities is that financial instruments do not share identical trading schedules. While stock exchanges operate on designated trading days, cryptocurrencies can trade continuously, including weekends. FinFolio currently aligns assets using common available observations and applies a 252-session annualization convention.

This provides a consistent basis for portfolio-level analysis, but it does not capture every price movement of continuously traded assets. For instance, the system may record the net price change between Friday and Monday while missing a temporary but substantial weekend drawdown.

Third, asset classifications do not always capture the full economic exposure of investment products.

Asset classifications often fail to capture true underlying economic exposures because multi-dimensional attribution frameworks rely on instrument-level data rather than decomposing fund holdings. For instance, broad equity ETFs are treated as single instruments instead of being broken down by their constituent stocks, while commodity products like USO track oil futures-whose returns diverge from spot crude prices due to their investment structure.

Lesson learned



Theory into Practice

Applied portfolio theory, risk-return analysis, and optimization to real market data, bridging classroom knowledge with practical financial applications.



Ownership & Problem-Solving

Developed independent thinking, decision-making, and prioritization skills while addressing real project challenges and constraints.



Teamwork & Communication

Learned to translate financial concepts into clear requirements and collaborate effectively across different project roles.

Declaration

I confirm that this report reflects the work that I am prepared to explain, calculate, demonstrate, or reproduce in Finfolio 2.0. Shared project artifacts are identified as collaborative where applicable, and my individual accountability is defined by the evidence presented in this report.

Field	Information
Full name	Hoang Khanh Linh
Student ID	2412380024
Signature	
Date	25/10/2026